Khai Nguyen

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Education

Northeastern University, Boston, MA — Expected Graduation: April 2030
Bachelor of Science in Mechanical Engineering and Computer Science
Relevant Coursework: Cornerstone of Engineering, Physics I-II for Engineering I, Calculus I-III, Differential Equations and Linear Algebra for Engineering, Statistics

Projects

Ledger–Savings Goal Tracker | React, JavaScript, Vite, CSS

- Built Ledger, a savings tracker that turns a target amount and deadline into a fixed weekly benchmark schedule, then measures logged deposits and withdrawals against it to surface ahead/behind pace, required weekly rate, and projected final total through a live chart and weekly check-in ledger persisted to localStorage.
- Isolated every money and date calculation into a single module containing no React or DOM code, making the core logic unit-testable in plain Node, and routed all persistence through one storage wrapper so a real backend could replace localStorage without touching the UI.
- Wrote 32 unit tests pinning down the edge cases that break naive implementations–already-passed deadlines, plan durations that don't divide evenly into weeks, withdrawals that would overdraw the balance, and floating-point drift on cumulative totals–and automated releases with a GitHub Actions pipeline that fails loudly rather than publishing a broken artifact to GitHub Pages.
- **Github:** Repository | **Live Demo:** ViewApp

Emberdeep | HTML, CSS, JavaScript

- Built Emberdeep, a browser-based roguelike dungeon crawler in vanilla JavaScript, HTML5 Canvas, and CSS–fully self-contained with zero external dependencies or build tools.
- Implemented procedural dungeon generation and BFS-based enemy AI, reusing the same graph-search algorithm for pathfinding and for validating that every randomly generated level is solvable (with automatic retries and a safe fallback).
- Wrote a headless Node.js test harness to stress-test dungeon generation and AI logic across hundreds of simulated runs, catching edge cases like unreachable exits and invalid spawns without a browser environment.
- **Github:** Repository | **Live Demo:** PlayGame

Rubik's Cube Speed Timer | HTML, CSS, JavaScript

- Built and deployed a responsive speedcubing timer web app using vanilla HTML, CSS, and JavaScript, implementing a competition-style state machine and performance.now()/requestAnimationFrame for millisecond-accurate timing with zero external dependencies.
- Designed a WCA-style scramble generator and statistics engine (Ao5, Ao12, session averages) with correct handling of edge cases like DNF penalties and insufficient sample sizes, validated through 50+ automated unit and integration tests.
- Implemented client-side data persistence with localStorage, including defensive parsing to gracefully recover from corrupted or missing data, and deployed the app via GitHub Pages.
- **Github:** Repository | **Live Demo:** PlayGame

Reaction Time Game | HTML, CSS, JavaScript

- Built and deployed an interactive browser-based reaction-time game featuring randomized timing, multiple difficulty levels, performance statistics, and persistent high scores using local storage.
- Developed game-state management and event-driven interactions in vanilla JavaScript to track user input and accurately measure reaction times.
- **Github**: Repository | **Live Demo:** Play Game

Skills

Technical: *(proficient)*: Java, Python, (*familiar)* JavaScript
Language: Vietnamese